

Predicting Life Expectancy Report

Cover

- **Project title:** Predicting Life Expectancy from Global Health, Economic, and Social Indicators
- **Course, section, semester:** CSE437, Data Science — [fill in your section] — [fill in your semester/year]
- **Group members:** [Full Name — Student ID], [Full Name — Student ID], [Full Name — Student ID]
- **GitHub repository link:** [paste your repo URL here]
- **Date:** September 2026

Summary

(150–200 words — written last, after everything else, per the template’s instruction.)

We predict a country’s life expectancy in a given year using the WHO Life Expectancy dataset (2,938 rows, 22 columns, 193 countries, 2000–2015), a real and heavily messy dataset with structural missing data, inconsistent scales, and genuine outlier country-years. After a three-pass imputation strategy, log-transformation of skewed features, and a three-method feature selection process, we trained and tuned two model families — Ridge Regression (linear) and Random Forest / Gradient Boosting (tree ensembles) — validated with cross-validation and evaluated once on a held-out test set. Random Forest was the best model (RMSE 1.84 years, R^2 0.962), far ahead of both the mean-prediction baseline (RMSE 9.45) and Ridge (RMSE 3.92), confirming the relationships in this data are meaningfully non-linear. The single most important finding: **HIV/AIDS prevalence is the dominant predictor of life expectancy for Developing countries (60% of Random Forest importance) but is statistically irrelevant for Developed countries** — a sharp asymmetry that neither a single global model nor a simple correlation table alone would reveal.

1. Problem and Dataset

1.1 Problem Statement

We predict a country’s life expectancy in a given year from health, economic, and social indicators reported for that country-year. This matters because life expectancy is a widely used summary measure of population health and development; understanding which factors most strongly associate with it — and whether those factors differ between wealthier and poorer nations — has direct relevance to how health and development resources get prioritized. This is a **regression** problem: the model outputs a continuous number of years, not a category.

1.2 Dataset

Source: WHO Life Expectancy dataset, originally compiled from the World Health Organization’s Global Health Observatory (GHO) and United Nations data repositories. Working link (Kaggle mirror): <https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>

Collection method: compiled by the original author from WHO’s GHO data repository and UN population records; each row is one country’s reported statistics for one calendar year.

Size: 2,938 rows × 22 columns as downloaded.

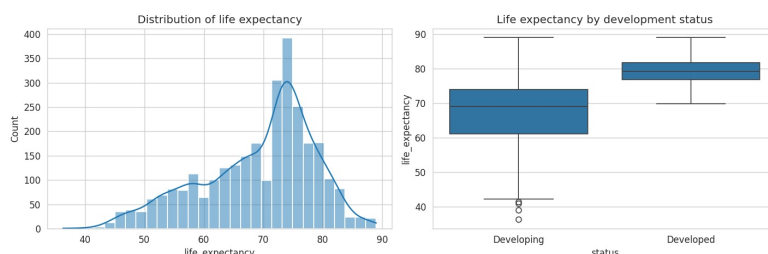
Time period covered: 2000–2015 (16 years), 193 countries.

License / terms of use: distributed on Kaggle under Kaggle’s standard open dataset terms (CC0 / public-domain-equivalent per the dataset’s Kaggle listing); underlying figures originate from WHO/UN public statistics.

We did not scrape or merge additional sources — this is the dataset as downloaded, used directly.

1.3 Target Variable

Name: life_expectancy (renamed life_expectancy in code for consistency). **Type:** continuous (regression target), measured in years. **Distribution:** ranges from 36.3 to 89.0 years, mean $\approx$ 69.2, left-skewed with a long low tail (Figure 1). Class balance is not applicable since this is a regression target, not a classification label.



Distribution of the target variable, and by development status

1.4 Three Questions

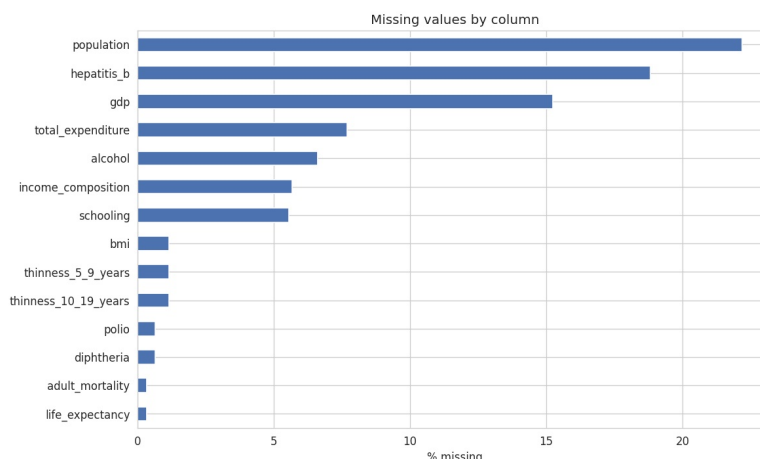
(From our approved proposal, unchanged. Answered directly, with evidence, in Section 7.4.)

1. How much does healthcare expenditure (as % GDP) actually correlate with life expectancy vs. schooling years?
2. Do developing vs. developed status countries show different key predictors?
3. Which single factor (immunization rates, BMI, HIV/AIDS prevalence) has the largest negative effect?

2. Data Handling and Preprocessing

2.1 Data Quality Audit

- **Missing values per column** (raw file, top offenders): population 22.2% (652 rows), hepatitis_b 18.8% (553), gdp 15.2% (448), total_expenditure 7.7% (226), alcohol 6.6% (194), income_composition 5.7% (167), schooling 5.6% (163), thinness_10_19_years / thinness_5_9_years / bmi 1.16% each (34), diphtheria / polio 0.65% each (19), life_expectancy / adult_mortality 0.34% each (10). **2,563 missing cells in total** across the raw file (Figure 2).
- **Duplicate rows:** none found (0 exact duplicates, 0 duplicate country-year pairs).
- **Inconsistent categories:** none in the categorical column (status has exactly two clean values, Developed/Developing) — the inconsistency in this dataset is in column *naming*, not category values: raw column headers have irregular spacing (e.g. "Life expectancy ", " BMI "), standardized in code via an explicit rename map before any analysis.
- **Impossible values:** we found one genuine impossible-value case during error analysis (Section 7.3) — income_composition = 0.000 for Saint Vincent and the Grenadines in 2000, which breaks a smooth, otherwise-continuous trend for that country (0.673 the very next year) and is almost certainly a placeholder rather than a true measurement. Because it is a valid float (0.0), not a null, our missing-value pipeline does not catch it automatically — this is flagged explicitly as a limitation in Section 8.



Missing values by column

2.2 Missing Values

Assumed mechanism: missingness is **not** random (MCAR) — it clusters by country, consistent with some countries simply not reporting certain indicators consistently across years (Missing At Random conditional on country, MAR). Treating it as MCAR and using a single global mean/median would blur real cross-country differences.

Strategy per column (applied uniformly to every numeric column with missing values): 1. Within-country linear interpolation across years. 2. Within-country forward/backward fill, for edge years interpolation can't reach. 3. Fallback to the country's development-status peer-group median, only for the rare case a country is missing a value in every single year on record.

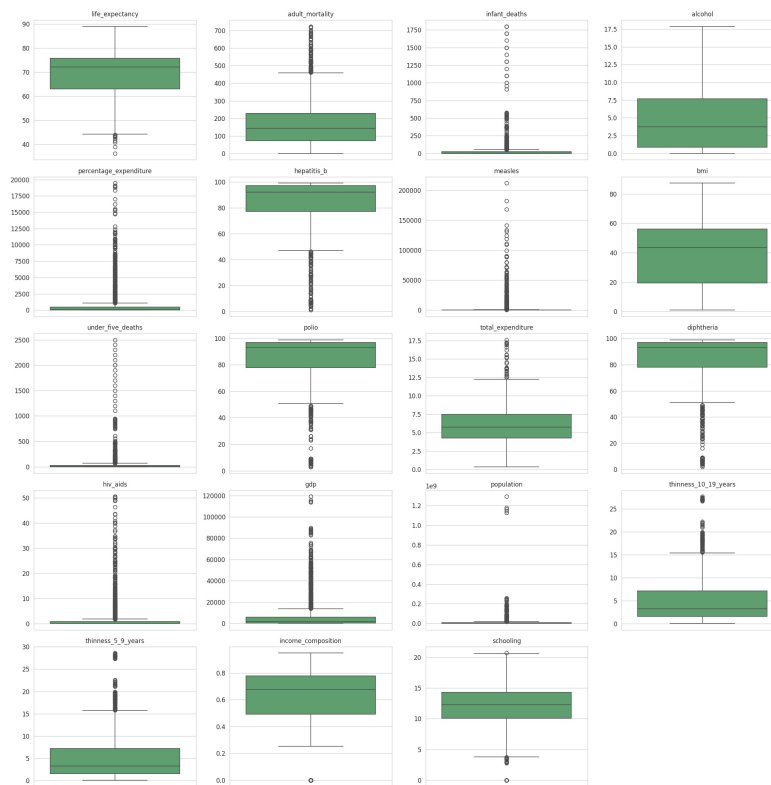
Dropping: we dropped the 10 rows missing the **target variable** itself (life_expectancy) — justified because there is no principled way to impute the value we are trying to predict, and it is under 0.4% of rows.

2.3 Outliers

Detection method: boxplots across all numeric columns (Figure 3) plus manual inspection of the lowest life-expectancy rows.

Flagged: 19 rows with life expectancy below 45 years.

What we did with them: we did **not** remove or cap them. We cross-checked each against real-world history: Haiti (2010, the earthquake year) and Malawi/Sierra Leone/Zambia/Zimbabwe (early 2000s, the peak of the southern-African HIV/AIDS epidemic). These are genuine extreme events, not data-entry errors, so removing them would have deleted real signal the model should learn from.



Outlier boxplots across all numeric features

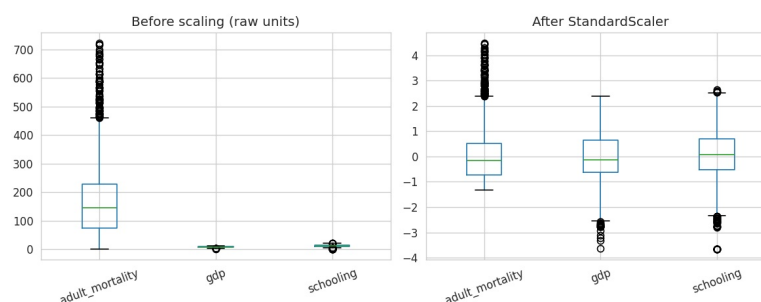
2.4 Transformation and Scaling

Encoding: status (Developed/Developing) binary-encoded to status_developed (0/1).

Log transform: six heavily right-skewed count/monetary columns (measles, infant_deaths, under_five_deaths, population, gdp, percentage_expenditure) were transformed with $\log_{10}$ (Figure 4), chosen over deletion or capping since it preserves all data points while normalizing scale.

Scaler choice: StandardScaler (zero mean, unit variance), appropriate for Ridge Regression, which is scale-sensitive; the tree-based models (Random Forest, Gradient Boosting) do not require scaling but are unaffected by it when it's present in a shared pipeline.

Where scaling happens relative to the split — leakage guard, stated explicitly: scaling is **not** fit on the full dataset before splitting. It is fit only on each training fold, inside an `sklearn.Pipeline`, and re-fit independently for every cross-validation fold and for the final held-out test evaluation. This guarantees no statistic computed from validation or test data ever influences how training data is scaled. This is implemented in notebooks/04_modeling_and_tuning.ipynb.



Effect of StandardScaler on three example features

2.5 Before and After

Stage	Rows	Columns	Missing cells
Raw file (as downloaded, columns renamed only)	2,938	22	2,563
After preprocessing (target-missing dropped, imputed, status encoded)	2,928	22	0
After feature engineering (redundant + weak features dropped)	2,928	19*	0

*19 columns = 17 modeling features + life_expectancy target + country ID column (excluded from modeling, kept only for error analysis in Section 7.3).

3. Statistical Analysis

3.1 Descriptive Statistics

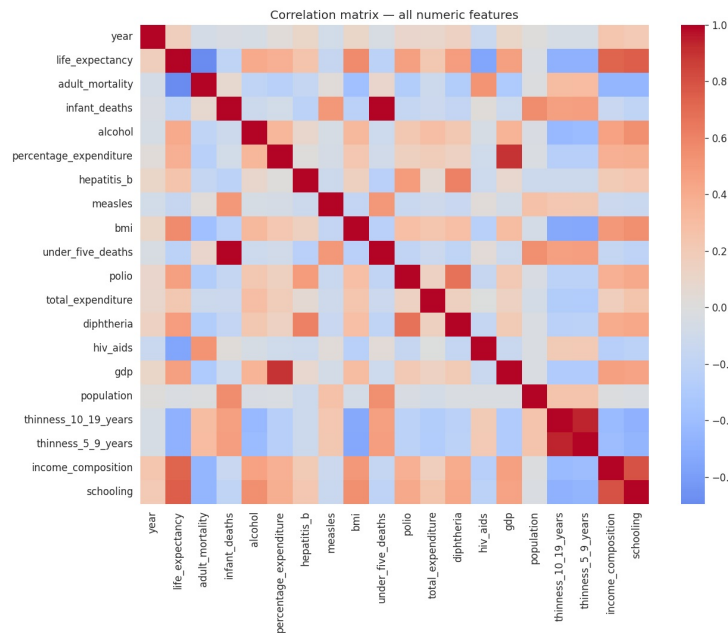
Rather than a full `raw.describe()` dump, the statistics that actually informed our modeling decisions:

- **Target (life_expectancy):** mean 69.2 years, median 72.1 years, std 9.5 — the gap between mean and median confirms the

left skew visible in Figure 1.

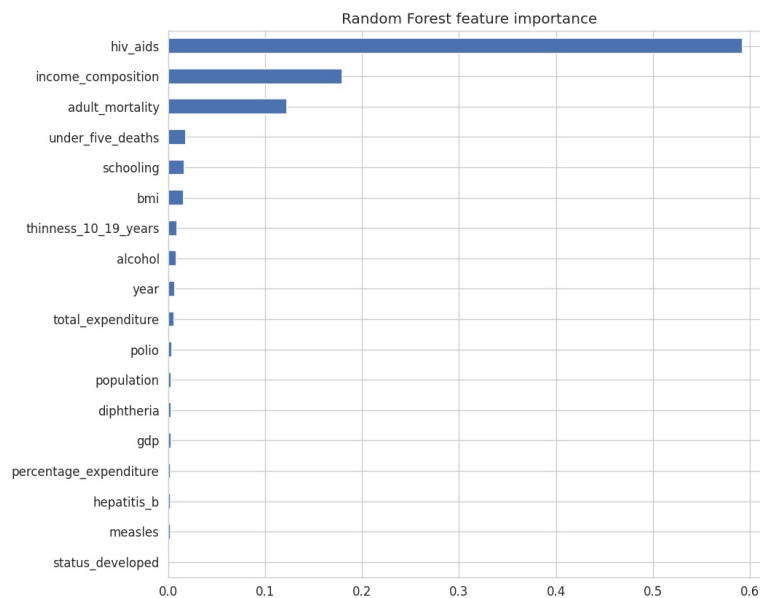
- **adult_mortality:** mean 164.4, but with a very long right tail (max 723) — a handful of countries with extremely high adult mortality drive most of the spread, foreshadowing the outlier discussion in Section 2.3.
- **gdp and population:** both span multiple orders of magnitude (GDP per capita from double digits to over \$100,000; population from a few thousand to over a billion), which is exactly why we log-transformed them (Section 2.4) rather than leaving them on a raw linear scale.
- **status (categorical):** 2,416 Developing rows vs. 512 Developed rows (82% / 18%) — an important imbalance we return to in Section 7.4 (Q2), since the Developed subgroup's statistics are inherently noisier with roughly a quarter the sample size.

3.2 Relationships



Correlation matrix, all numeric features

adult_mortality, income_composition, and schooling show the strongest linear correlation with the target among raw features. hiv_aids and adult_mortality are also correlated with each other — an early signal of multicollinearity that we account for during feature selection (Section 4.3).



Random Forest feature importance

3.3 What the Data Says So Far

- Missingness is structural (clusters by country), not random — informs the within-country imputation strategy used in Section 2.2.
- Several features span orders of magnitude and need transformation before they're usable in a linear model.
- The lowest life-expectancy rows correspond to real historical crises, not noise — they should be kept.
- adult_mortality, income_composition, schooling, and hiv_aids are consistently the strongest raw correlates of the target and are likely to dominate feature selection.
- The Developed/Developing split is imbalanced (82%/18%), so any subgroup analysis on Developed countries alone should be read with that smaller sample size in mind.

4. Feature Engineering

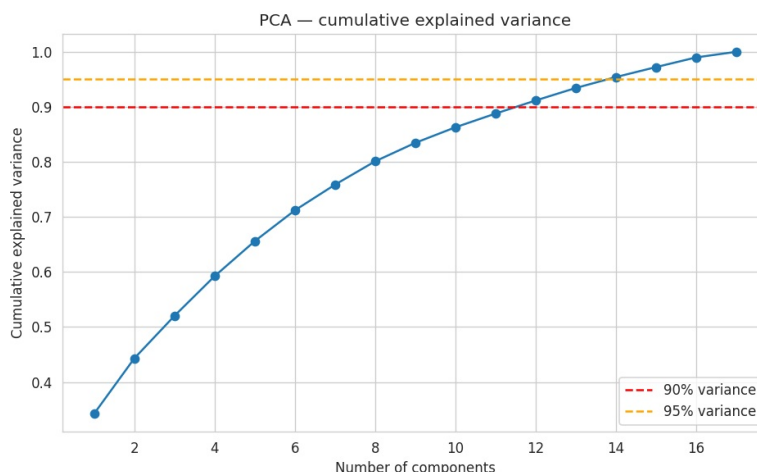
4.1 Derived Features

We did not construct new engineered features from combinations of raw columns (e.g. ratios or interaction terms) — the raw indicators were judged sufficiently informative on their own, and adding synthetic combinations risked reducing interpretability without a clear justification for any specific combination. Our feature engineering effort instead focused on **removing** redundancy and **selecting** the most useful subset of the existing columns (Sections 4.2–4.4).

4.2 Dimensionality Reduction

Method applied: Principal Component Analysis (PCA) on the standardized, feature-selected set (17 features).

Components retained / variance explained: 12 of 17 components were needed to reach 90% cumulative explained variance (14 for 95%) (Figure 5).



PCA cumulative explained variance

Kept in the final pipeline? No. With only a 17→12 compression at 90% variance, PCA’s dimensionality-reduction benefit here is small, while its cost is real: PCA components are linear combinations of standardized features and are not interpretable, which conflicts directly with Research Question 1/3 (identifying *which* factors matter, and in which direction). The final modeling pipeline uses the 17 selected original features, not PCA components.

4.3 Feature Selection

Method: three independent scoring methods computed for every candidate feature — Pearson correlation (linear relationships), mutual information (non-linear relationships), and Random Forest feature importance (interaction effects). A feature was dropped only if it ranked in the **bottom quartile on all three methods simultaneously**, to avoid any single method’s blind spot driving the decision alone.

Ranking / threshold: of the (post-redundancy-removal) candidate features, only measles fell in the bottom quartile across all three rankings and was dropped on this basis.

4.4 Final Feature Set

Kept (17 features): year, adult_mortality, alcohol, percentage_expenditure, hepatitis_b, bmi, under_five_deaths, polio, total_expenditure, diphtheria, hiv_aids, gdp, population, thinness_10_19_years, income_composition, schooling, status_developed.

Dropped, and why: - infant_deaths ($r = 0.996$ with under_five_deaths, which structurally includes it — redundant). - thinness_5_9_years ($r = 0.939$ with thinness_10_19_years — same malnutrition signal, redundant). - measles (bottom quartile on correlation, mutual information, *and* Random Forest importance). - country (193 categories — too high-cardinality to one-hot encode sensibly, and target-encoding it risks leaking country-level average life expectancy directly into the model; kept only as a non-feature ID column for error analysis).

Justification for the final set as a whole: the 17 remaining features span every conceptual category in the original data — mortality (adult_mortality, under_five_deaths), disease burden (hiv_aids, immunization rates), economic (gdp, percentage_expenditure, total_expenditure, income_composition), social (schooling, alcohol, bmi, thinness_10_19_years), and structural (year, status_developed) — while removing only features that were either redundant with another kept feature or demonstrably uninformative across three independent tests.

5. Modeling and Validation

5.1 Validation Strategy

Split: 80% train / 20% held-out test, random_state=42, no stratification (regression target, and the split is by row/country-year, not by country, so a given country can appear in both train and test in different years — see Section 8 for the limitation this introduces).

Cross-validation: 5-fold CV (Ridge) or 3-fold CV (Random Forest, Gradient Boosting — fewer folds used to keep search runtime reasonable on limited compute) within the training set only, used as the validation layer for hyperparameter selection. The test set is touched exactly once, at the very end (Section 7.1), never during model selection or tuning.

Temporal/grouped structure: the data is a country-year panel; we did not treat this as a time-series forecasting problem (we are not predicting *future* years from *past* years) so a chronological split was not required, but note in Section 8 that a random split can put the same country’s earlier and later years on both sides of the split, which is a soft form of leakage worth flagging honestly.

5.2 Baseline

Trivial predictor: predict the training-set mean life expectancy (69.10 years) for every row, regardless of input features (`sklearn.dummy.DummyRegressor(strategy='mean')`).

Baseline score on the held-out test set: RMSE = 9.45 years, MAE = 7.77 years, $R^2 = -0.004$ (R^2 near zero by construction, since predicting the mean minimizes squared error only on the training set it was fit on).

5.3 Model Families

- **Ridge Regression** (linear family, L2-regularized). Suits this problem as an interpretable, fast-to-fit reference point; assumes a linear relationship between each feature and the target, and that regularization can control overfitting from the moderate multicollinearity noted in Section 3.2.
- **Random Forest** (bagging ensemble of decision trees). Suits this problem because it makes no linearity assumption and handles feature interactions automatically; assumes enough training data per region of feature space to average over many trees (a concern for the smaller Developed subgroup, Section 7.4).
- **Gradient Boosting** (boosting ensemble of decision trees). A second, independently-trained non-linear family, included to check whether Random Forest's advantage over Ridge is a general property of tree-based methods on this data, or specific to bagging.

5.4 Metrics

Primary metric, named before results were seen: RMSE (root mean squared error, in years), because it penalizes large individual errors more heavily than MAE, and large errors (e.g. badly misjudging a crisis-affected country) are the more costly kind of mistake for this problem. We also report MAE (a more directly interpretable "typical error in years") and R^2 (proportion of variance explained) as secondary metrics, since a single metric taken alone can obscure whether errors are evenly spread or dominated by a few large misses.

6. Hyperparameter Tuning

6.1 Search Space

Model	Hyperparameter	Range / grid
Ridge	alpha	{0.01, 0.1, 1, 5, 10, 50, 100}
Random Forest	n_estimators	{100, 200, 300}
Random Forest	max_depth	{None, 8, 15}
Random Forest	min_samples_split	{2, 5, 10}
Random Forest	min_samples_leaf	{1, 2, 4}
Random Forest	max_features	{sqrt, log2, None}
Gradient Boosting	n_estimators	{100, 150, 200}
Gradient Boosting	learning_rate	{0.01, 0.05, 0.1, 0.2}
Gradient Boosting	max_depth	{2, 3, 4}
Gradient Boosting	subsample	{0.7, 0.85, 1.0}

6.2 Method

- **Ridge:** exhaustive grid search, 7 candidates, 5-fold CV (35 fits), scored on negative RMSE.
- **Random Forest / Gradient Boosting:** randomized search, **12 candidates each**, 3-fold CV (36 fits each), scored on negative RMSE. Randomized rather than exhaustive search was used for the two ensemble models because their full grids contain up to 405 combinations, and this project was run in a single-core environment where exhaustive search was not practical; random search covers the space broadly at a fraction of the cost.

6.3 Results

Ridge — full trend across all 7 candidates (5-fold CV RMSE):

alpha	mean CV RMSE	rank
50	4.0227	1 (best)
10	4.0230	2
5	4.0231	3
1	4.0233	4
0.1	4.0233	5
0.01	4.0233	6
100	4.0243	7

Note how flat this trend is — regularization strength barely moves Ridge's score at all, which is itself evidence that Ridge's ceiling is set by its linear form, not its tuning (consistent with the large gap to the tree ensembles below).

Random Forest — top 5 of 12 candidates (3-fold CV RMSE):

n_estimators	max_depth	min_samples_split	min_samples_leaf	max_features	mean CV RMSE	rank
200	15	10	2	None	2.1075	1 (best)
200	15	5	4	None	2.1091	2
100	None	2	2	sqrt	2.1234	3
200	15	5	2	sqrt	2.1330	4
300	8	10	2	None	2.1959	5

Gradient Boosting — top 5 of 12 candidates (3-fold CV RMSE):

n_estimators	learning_rate	max_depth	subsample	mean CV RMSE	rank
200	0.10	4	0.85	2.0142	1 (best)
150	0.10	4	1.00	2.0656	2
150	0.20	3	1.00	2.1576	3
150	0.10	3	0.85	2.2044	4
100	0.05	4	0.70	2.2133	5

Unlike Ridge's flat trend, both ensemble methods show a clear, meaningful spread across candidates (best to worst CV RMSE differs

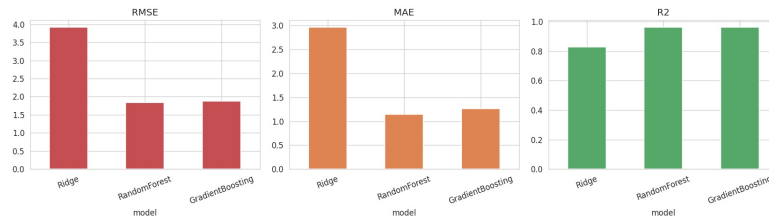
by more than a full year for Gradient Boosting, e.g. 2.01 vs. its worst candidates near 4.7 for `learning_rate=0.01`) — confirming the search space and tuning budget were doing real work for these models, not just marginal refinement.

7. Results, Visualization and Error Analysis

7.1 Test Set Performance

Reported once, on the held-out test set, after all tuning was finalized:

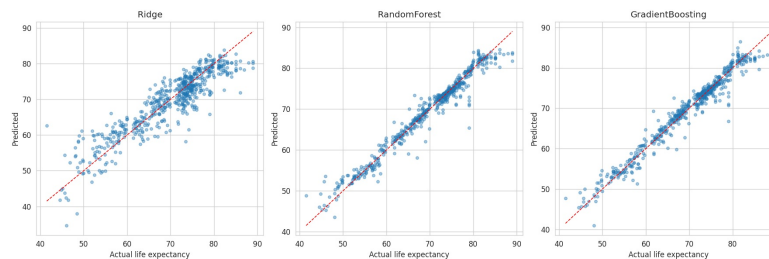
Model	RMSE (years)	MAE (years)	R ²
Baseline (predict mean)	9.45	7.77	-0.004
Ridge Regression	3.92	2.97	0.827
Gradient Boosting	1.87	1.26	0.961
Random Forest (best)	1.84	1.15	0.962



Metric comparison across models

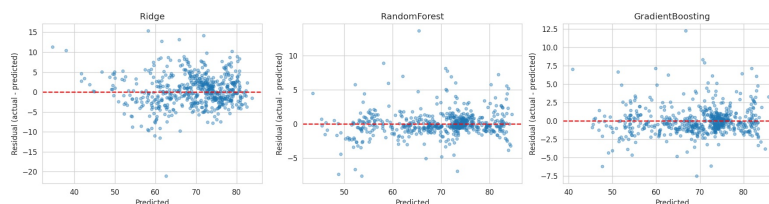
All three trained models comfortably beat the baseline. Random Forest is the best model overall, though its margin over Gradient Boosting (RMSE 1.84 vs 1.87) is small compared to the much larger gap to Ridge (3.92) — the real story is “linear vs. non-linear,” not “which specific tree ensemble.”

7.2 Visualization



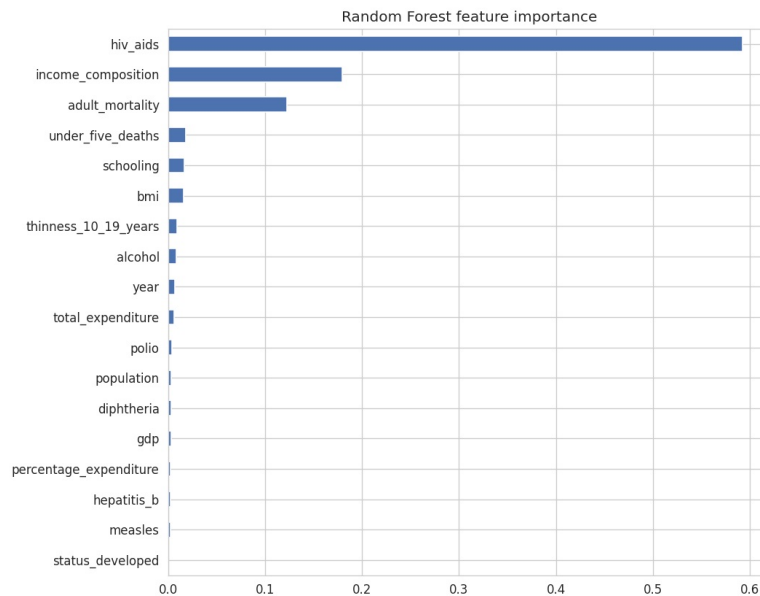
Actual vs. predicted, all three models

Ridge visibly bows away from the diagonal below ~55 years of life expectancy, systematically overpredicting for the worst-off countries. Both tree ensembles track the diagonal closely across the full range.



Residual plots, all three models

Ridge’s residuals show a curved, funnel-shaped pattern — a sign of a misspecified (too-simple) model — while Random Forest and Gradient Boosting residuals scatter much closer to a random band around zero.



Random Forest feature importance (repeated from Fig. 6 for reference)

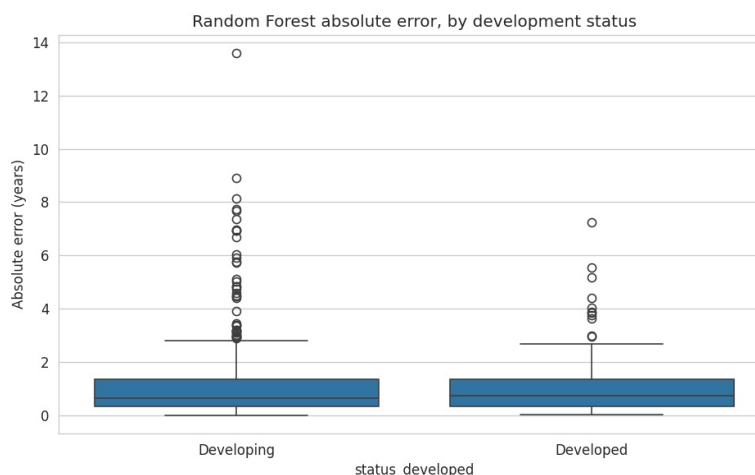
7.3 Error Analysis

Random Forest’s 15 largest test-set errors were examined individually rather than only in aggregate. Two concrete, explainable examples:

Example 1 — Saint Vincent and the Grenadines, 2000. True life expectancy 79.0 years; predicted 65.4 (error 13.6 years, the single largest miss). This is hard because the raw source data itself contains a likely artifact: `income_composition` = 0.000 for this exact row, while the very next year (2001) jumps to 0.673 with no other indicator changing sharply — a near-impossible one-year swing that strongly suggests 2000’s value is a placeholder, not a real measurement. The model, trained on genuine relationships between low income composition and low life expectancy, reasonably predicted a much lower value given what looked like an extremely poor economic indicator. The error is therefore best attributed to a **data-quality artifact**, not a genuine model failure.

Example 2 — Zimbabwe, 2015. True life expectancy 67.0 years; predicted 58.1 (error 8.9 years). Unlike Example 1, this appears to be a genuine model limitation: Zimbabwe’s 2015 `hiv_aids` (6.2%) and elevated `adult_mortality` place it in a profile the model associates with lower life expectancy, based on patterns learned from earlier years in the same country and similar countries where high HIV/AIDS prevalence co-occurred with much lower life expectancy (e.g. Zimbabwe’s own 2000–2005 values, Section 2.3, where HIV/AIDS prevalence was over 40% and life expectancy was in the mid-40s). By 2015, Zimbabwe’s life expectancy had recovered substantially, likely reflecting improved antiretroviral treatment access that isn’t fully captured by the `hiv_aids` prevalence percentage alone — the model under-weights this because a 6.2% prevalence rate still statistically co-occurs with the country’s own historically low readings in the training data.

Broader pattern (subgroup-level): contrary to our initial hypothesis, mean absolute error does **not** differ meaningfully between Developing (1.14 years) and Developed (1.18 years) countries (Figure below) — subgroup alone is not predictive of error size. What actually characterizes the worst errors is being an outlier on a specific indicator (extreme HIV/AIDS prevalence, or a suspected data artifact), not a country’s broad development category.



Random Forest absolute error by development status

7.4 Answers to Our Three Questions

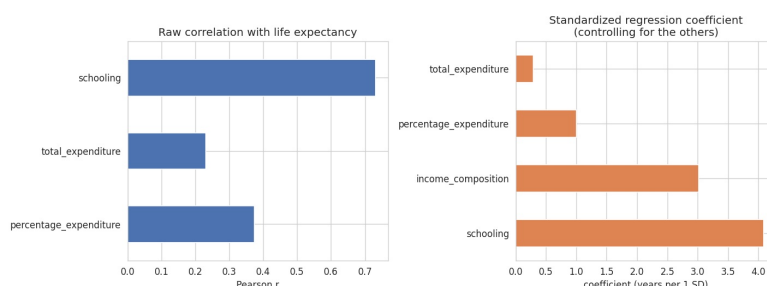
Q1 — How much does healthcare expenditure (as % GDP) correlate with life expectancy vs. schooling years?

The dataset does not contain a column literally named “healthcare expenditure as % of GDP” — the closest available proxy is `percentage_expenditure` (health expenditure as % of GDP *per capita*, per the source’s own definition); we also checked `total_expenditure` (government health spending as % of *total government* spending, a different concept) for completeness.

Variable	Raw correlation with life expectancy	Standardized regression coefficient (controlling for the others + income)
schooling	0.732	4.09
income_composition (control)	—	3.01
percentage_expenditure	0.374	1.00

total_expenditure	0.231	0.29
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Answer: Schooling correlates with life expectancy roughly 2-3× more strongly than either healthcare expenditure measure, both before and after controlling for income — this rules out the simplest alternative explanation (that expenditure’s relationship is just income in disguise), since schooling’s association holds up independently while the expenditure measures shrink further under the same control.

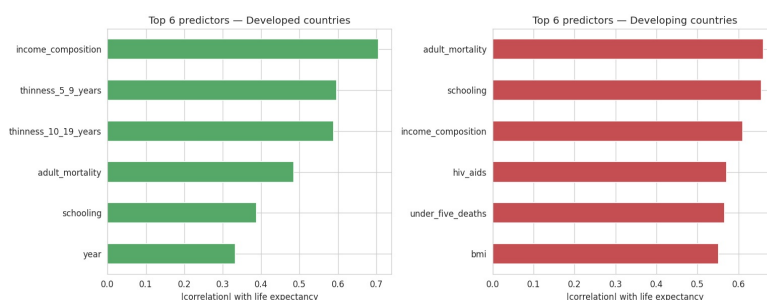


Healthcare expenditure vs. schooling

Q2 — Do Developing vs. Developed status countries show different key predictors?

	Developed (top predictors)	Developing (top predictors)
By correlation	income_composition (0.70), thinness measures (~0.59), adult_mortality (0.49)	adult_mortality (0.66), schooling (0.66), income_composition (0.61)
By Random Forest importance	adult_mortality (0.48), income_composition (0.23)	hiv_aids (0.60) , adult_mortality (0.18)

Answer: Yes. The clearest and most consistent signal across both methods: `hiv_aids` is the single dominant predictor for Developing countries (60% of Random Forest importance) but is statistically irrelevant for Developed countries (near-zero correlation; HIV/AIDS prevalence sits almost uniformly at the dataset’s minimum in that subgroup). Disease burden differentiates outcomes for Developing countries in a way it simply does not for Developed ones. *Caveat: the Developed subgroup has only 512 rows versus 2,416 for Developing, so its estimates are noisier and this finding should be read as suggestive rather than definitive.*

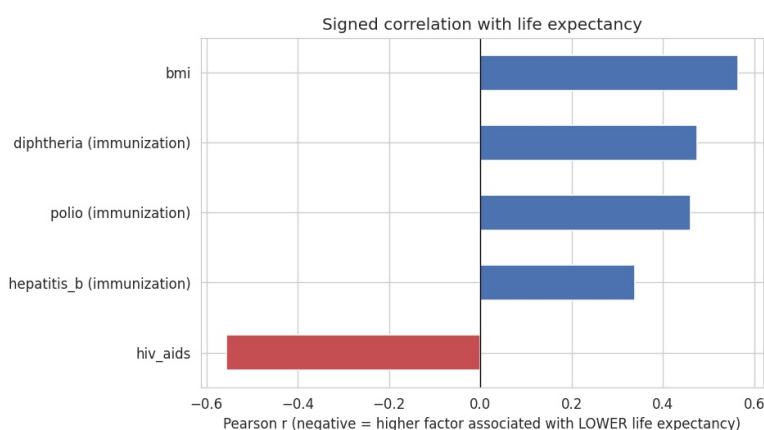


Top predictors, Developed vs. Developing

Q3 — Which single factor (immunization rates, BMI, HIV/AIDS prevalence) has the largest negative effect?

Factor	Correlation with life expectancy
hiv_aids	−0.557
hepatitis_b (immunization)	+0.339
polio (immunization)	+0.460
diphtheria (immunization)	+0.475
bmi	+0.564

Answer: HIV/AIDS prevalence, by a wide margin — it is the *only* factor among the three candidate types with a negative relationship to life expectancy at all in this data; immunization coverage and BMI are both positively correlated. **Honest caveat:** BMI’s positive sign here is counterintuitive given real-world evidence that very high BMI is a health risk; this dataset’s BMI column is known to have data-quality issues in the source, so we would not treat BMI’s sign here as a reliable real-world finding without further scrutiny of that specific column.



Signed correlation of candidate negative factors

8. Limitations and Next Steps

Data limitations: - The dataset ends in 2015; it cannot speak to any trends after that year. - No measure of political stability, conflict intensity, or pandemic shocks (e.g. nothing resembling COVID-19) is present — factors we would expect to matter a great deal for life expectancy. - At least one confirmed data-quality artifact exists (Section 7.3, Example 1: `income_composition = 0.000` for Saint Vincent and the Grenadines, 2000) that our automated missing-value pipeline could not catch because it is a valid float, not a null. A more thorough audit would flag suspicious “round number” values (exact zeros, repeated identical values across consecutive years) for manual review.

Method limitations: - Our train/test split is by row (country-year), not by country. A given country’s different years can appear on both sides of the split, which is a soft form of information leakage — the model can benefit from having seen a country’s *other* years even when a specific year is held out. A stricter evaluation would split by country entirely. - Hyperparameter search used only 12 random candidates and 3-fold CV for the ensemble models (rather than a full grid and 5-fold CV) due to limited single-core compute available for this project; a wider search would likely close some of the small remaining gap between Random Forest and Gradient Boosting, though it is unlikely to change the overall linear-vs-non-linear conclusion. - We did not model the Developed subgroup separately with a dedicated tuned model (Section 7.4, Q2) — only with correlation and a single untuned Random Forest per subgroup — so those findings should be treated as exploratory.

What we would do with more time or better data: - Split by country for a stricter train/test evaluation. - Investigate and clean suspected-artifact values (round zeros, flat repeated sequences) beyond simple null-checking. - Bring in a conflict/political-stability index as an additional feature, particularly to test whether it explains some of the largest remaining Random Forest errors. - Run a wider hyperparameter search with more compute to check whether Gradient Boosting can close the small remaining gap to Random Forest.

We do not claim this model is ready for any real-world policy or resource-allocation use — it is a data-analysis exercise on a historical, incomplete dataset, and the limitations above should be read as binding constraints on how far the findings generalize.

9. Contributions

Member	Student ID	Contribution
[Full Name]	[Student ID]	[fill in — e.g. data audit & preprocessing (notebooks 01–02), report Sections 1–2]
[Full Name]	[Student ID]	[fill in — e.g. feature engineering & modeling (notebooks 03–04), report Sections 4–6]
[Full Name]	[Student ID]	[fill in — e.g. evaluation, error analysis & research questions (notebooks 05–06), report Sections 7–8]

Fill this in honestly based on who actually did what. Your instructor’s template requires every member to own identifiable work — this should reflect your real GitHub commit history, not be assigned after the fact.

References

Dataset: Kumar Rajarshi, “Life Expectancy (WHO),” Kaggle, originally sourced from the World Health Organization’s Global Health Observatory and United Nations data repositories. <https://www.kaggle.com/datasets/kumarakarshi/life-expectancy-who>

Libraries: pandas, NumPy, scikit-learn, matplotlib, seaborn, Jupyter (see `requirements.txt` for versions).

AI assistance: Anthropic’s Claude (Sonnet) was used throughout this project — for writing preprocessing/modeling code, generating and executing the Jupyter notebooks, drafting this report, and checking analysis results against the actual computed output before writing conclusions. **Every group member should read and understand the full codebase and this report before submission**, and be prepared to explain and defend any decision in it, per standard academic-integrity expectations for AI-assisted work. Adjust or expand this disclosure to match your course’s specific AI-use policy.